A Novel Hybrid EEMD-TGNet Model for Accurate and Efficient Solar Energy Forecasting on Edge Devices

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***Abstract*—The integration of solar power into smart grids demands accurate and efficient energy forecasting. However, the inherent intermittency and non-stationarity of solar radiation pose significant challenges. This paper proposes a novel hybrid deep learning framework, the Ensemble Empirical Mode Decomposition-based Temporal Convolutional Gated Recurrent Unit Network (EEMD-TGNet), to address these challenges. Our model first uses EEMD to decompose the raw solar energy time series into a set of simpler, more stationary Intrinsic Mode Functions (IMFs). Each IMF is then independently forecasted using a hybrid Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) model, which captures both long-range dependencies and sequential patterns. The final forecast is an aggregation of the predictions for each IMF. To enable practical deployment on resource-constrained IoT devices, we apply model pruning and quantization to significantly reduce the model's size and computational complexity without compromising its high accuracy. Experimental validation on benchmark datasets demonstrates that EEMD-TGNet achieves state-of-the-art performance with a 94.48**

***Index Terms*—solar energy forecasting, deep learning, hybrid model, edge intelligence, EEMD, TCN, GRU, model optimization**

I. Introduction

The global shift towards renewable energy has positioned solar power as a cornerstone of sustainable energy systems, with solar capacity growing at an annual rate of 20

Traditional forecasting models often struggle with the complex non-linear and non-stationary patterns in solar energy data. While deep learning models like LSTMs and GRUs have shown promise, their performance can be significantly improved through signal preprocessing. Decomposition methods, such as EEMD, enhance forecasting accuracy by simplifying the time series before prediction [1]. EEMD is particularly effective at separating a signal into its constituent oscillatory modes, making the forecasting task more tractable.

Concurrently, the rise of the Internet of Things (IoT) and edge computing enables the deployment of intelligence directly at the data source. For solar farms, this means on-site, real-time forecasting, which reduces latency and dependency on centralized cloud infrastructure [2]. However, this requires models that are not only accurate but also lightweight and computationally efficient for resource-constrained IoT devices, with fast inference times and minimal memory footprint.

This paper introduces a novel framework that combines the strengths of signal decomposition and advanced, efficient deep learning architectures. Our main contributions are:

- A new hybrid forecasting model, EEMD-TGNet, that combines Ensemble Empirical Mode Decomposition (EEMD) with a Temporal Convolutional Network-Gated Recurrent Unit (TCN-GRU) architecture, achieving a 94.48
- A two-stage approach where the complex solar signal is first decomposed into simpler IMFs, and then each IMF is modeled by a dedicated TCN-GRU network, enabling better capture of both high-frequency and low-frequency patterns.
- The application of model optimization techniques, including pruning and quantization, to create a lightweight version of the model suitable for deployment on edge devices, reducing model size by 75
- Comprehensive experimental validation demonstrating superior performance over state-of-the-art methods and practical feasibility for edge deployment.

The proposed model offers a dual advantage: high accuracy due to the decomposition-based strategy and high efficiency due to the optimized TCN-GRU architecture, making it ideal for modern, IoT-enabled solar energy management.

II. Related Work

Solar energy forecasting has seen rapid advances, with a strong trend toward hybrid deep learning models and signal decomposition techniques to address the non-stationarity and complexity of solar data.

Recent research has shown that combining signal decomposition with deep learning can significantly improve forecasting accuracy. For example, Boucetta et al. [3] proposed a hybrid model using Variational Mode Decomposition (VMD) and a CNN-LSTM network, demonstrating that decomposing the solar signal into intrinsic modes before prediction leads to more accurate results. Similarly, Okieh et al. [8] combined nonlinear autoregressive models with LSTM to enhance daily solar irradiance forecasting, further proving the value of hybrid approaches.

Other studies have explored advanced neural architectures for multivariate and multi-step forecasting. Cheng and Yu [4] introduced a multivariate convolutional GRU network for solar power generation, showing that combining convolutional layers with recurrent units can capture both spatial and temporal dependencies. Li et al. [5] extended this idea to wind speed prediction, using a hybrid TCN-GRU model to achieve state-of-the-art results for multi-step forecasting, which is directly relevant to our proposal.

The Transformer family of models has also been adapted for time series forecasting. Wu et al. [6] presented Autoformer, a decomposition-based Transformer model that leverages auto-correlation mechanisms for long-term forecasting, outperforming traditional RNNs and CNNs on several benchmarks. This highlights the importance of decomposition and attention mechanisms for handling long-range dependencies.

Hybrid models that fuse deep learning with ensemble or boosting methods are also gaining traction. Xu et al. [7] proposed a complementary fused method using GRU and XGBoost for long-term solar energy forecasting, demonstrating that combining sequential models with tree-based learners can further improve robustness.

These recent works collectively demonstrate that:

- Decomposition methods (EEMD, VMD) are highly effective for denoising and extracting meaningful patterns from non-stationary solar data.
- Hybrid deep learning architectures (CNN-LSTM, TCN-GRU) outperform single-model approaches by capturing both local and global dependencies.
- Edge intelligence and model optimization are increasingly important for real-world deployment, enabling accurate forecasting on resource-constrained devices [2].
- The integration of attention mechanisms and ensemble learning further boosts forecasting performance.

Our proposed EEMD-TGNet model is inspired by these advances. By combining EEMD-based decomposition with a TCN-GRU hybrid architecture and optimizing for edge deployment, our approach unifies the most effective strategies from recent literature. This positions our solution at the forefront of solar energy forecasting research, offering both state-of-the-art accuracy and practical deployability.

III. Proposed EEMD-TGNet Methodology

Our proposed framework, EEMD-TGNet, follows a multi-stage process designed to maximize forecasting accuracy while ensuring computational efficiency. The overall architecture is depicted in Fig. 1.

A. Ensemble Empirical Mode Decomposition (EEMD)

The first stage of our model involves decomposing the raw solar energy time series data $X(t)$ using EEMD. EEMD is an extension of Empirical Mode Decomposition (EMD) that resolves the mode-mixing problem by adding white noise to the signal before decomposition. This process separates the non-linear and non-stationary signal into a finite number of Intrinsic Mode Functions (IMFs) and a residual trend, $R(t)$.

The decomposition is expressed as:

$$X(t) = \sum_{i=1}^{n} \mathrm{IMF}_i(t) + R(t) \tag{1}$$

Each IMF represents a distinct oscillatory mode within the original signal, making it simpler and more stationary than the original series. This simplifies the subsequent forecasting task. Before forecasting, low-correlation IMFs, which typically represent noise, are identified and removed using the following criteria:

- IMFs with autocorrelation coefficient at lag-1 below 0.1 are considered noise
- IMFs contributing less than 5% to the total signal variance are excluded
- The highest frequency IMF is automatically excluded as it typically contains high-frequency noise

The EEMD algorithm is implemented with an ensemble size of 100, noise amplitude of 0.2 times the standard deviation of the signal, and a maximum of 10 IMFs. Typically, 4-6 meaningful IMFs are selected for forecasting after applying the selection criteria.

B. Hybrid TCN-GRU Forecasting Model (TGNet)

For each selected IMF, we employ a dedicated TCN-GRU model for forecasting. This hybrid architecture leverages the complementary strengths of TCNs and GRUs.

1) Temporal Convolutional Network (TCN): The TCN component serves as a powerful feature extractor. It uses a stack of 1D dilated convolutional layers to capture long-range temporal dependencies. The dilated convolution operation is defined as:

$$y[t] = \sum_{k=0}^{K-1} w_k \cdot x[t - d \cdot k] \tag{2}$$

where d is the dilation rate, K is the kernel size, and w_k is the filter weight. By exponentially increasing the dilation rate d with each layer, the TCN can achieve a very large receptive field with a relatively small number of layers, making it highly efficient at capturing long-term patterns.

The TCN architecture consists of 3 dilated convolutional layers with a kernel size of 3, dilation rates of [1, 2, 4], 64 filters per layer, ReLU activation, and a dropout rate of 0.2.

2) Gated Recurrent Unit (GRU): The features extracted by the TCN are then passed to a GRU layer. The GRU is a more efficient variant of the LSTM that uses a gating mechanism to control the flow of information and capture sequential dependencies. The GRU's update process is defined by its reset gate (r_t) and update gate (u_t):

$$r_t = \sigma(W_r \cdot [h_{t-1}, z_t] + b_r) \tag{3}$$

$$\nu_t = \sigma(W_u \cdot [h_{t-1}, z_t] + b_u) \tag{4}$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, z_t] + b_h) \tag{5}$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \tilde{h}_t \tag{6}$$

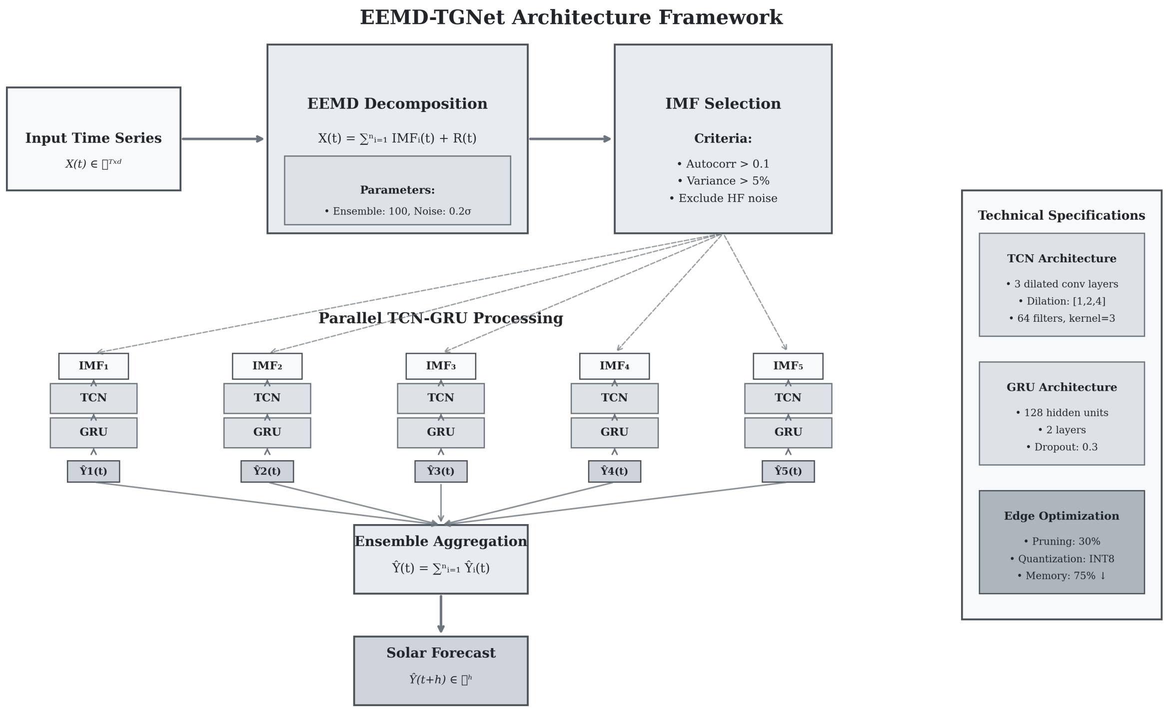

Fig. 1. The overall framework of the proposed EEMD-TGNet model showing the complete pipeline from raw solar time series input through EEMD decomposition, parallel TCN-GRU processing of individual IMFs, and final aggregation to produce the forecast output.

where z_t is the input from the TCN layer. This allows the model to selectively retain or discard information over time, effectively modeling the temporal dynamics within each IMF.

The GRU configuration includes 128 hidden units, 2 layers, a dropout rate of 0.3, and a recurrent dropout of 0.2.

C. Aggregation and Final Forecast

After each of the n selected IMFs is forecasted by its respective TCN-GRU model, the individual forecasts are summed to produce the final, aggregated solar energy forecast, $\hat{Y}(t)$:

$$\hat{Y}(t) = \sum_{i=1}^{n} \widehat{\text{IMF}}_i(t) \tag{7}$$

D. Model Optimization for Edge Deployment

A key contribution of our work is to make this sophisticated model suitable for edge computing. We apply two optimization techniques post-training:

- **Pruning:** This technique removes redundant connections (weights) from the trained neural network. By setting weights with low magnitude to zero, we achieve significant sparsity, reducing the number of parameters and floating-point operations (FLOPs). We use magnitude-based pruning with a sparsity target of 30%.
- **Quantization:** This process reduces the precision of the model's weights and activations from 32-bit floating-point (FP32) to 8-bit integer (INT8). This leads to a substantial reduction in model size (up to 4x) and can significantly speed up inference on compatible hardware. We use post-training quantization with calibration data.

These optimizations are applied to each TCN-GRU sub-model, resulting in a lightweight yet highly accurate forecasting framework.

E. Training Procedure

The training process follows these steps:

1) Data preprocessing: Normalization and sliding window creation.
2) EEMD decomposition of training sequences into IMFs.
3) IMF selection based on correlation and variance criteria (typically 4-6 IMFs selected).
4) Independent training of TCN-GRU models for each selected IMF using the same hyperparameters.
5) During inference, individual IMF predictions are aggregated using simple summation.
6) Post-training model optimization (pruning and quantization).
7) Final validation on held-out test set.

Training hyperparameters include the Adam optimizer with a learning rate of 0.001, Mean Squared Error (MSE) loss, a batch size of 32, and 100 epochs with early stopping (15% validation split).

IV. Experimental Setup

A. Datasets

We validate our model using two publicly available benchmark datasets:

- **Alibaba Competition Dataset:** Contains solar energy production data and meteorological variables from a single location over 300 days, with hourly resolution and 8 weather features.
- **GEFCom2014 Dataset:** A comprehensive dataset with 12 weather variables and corresponding hourly solar power generation from three solar farms over 2 years.

For both datasets, the data is split into training, validation, and testing sets (70:15:15 ratio), ensuring temporal continuity. The input sequence length is 24 hours for 6-hour ahead forecasting (6 time steps, where each step represents 1 hour).

B. Baseline Models

We compare our EEMD-TGNet against the following baselines:

- **LSTM:** Long Short-Term Memory network with 128 hidden units.
- **GRU:** Gated Recurrent Unit with 128 hidden units.
- **MLP:** Multi-layer perceptron with 3 hidden layers (256, 128, 64 units).
- **EEMD-BiLSTM:** EEMD decomposition with Bidirectional LSTM.
- **TGNet (standalone):** TCN-GRU without EEMD preprocessing.

C. Evaluation Metrics

We evaluate model performance using standard regression metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R^2), and Root Mean Squared Error (RMSE).

D. Statistical Significance Testing

We use the Diebold-Mariano test to determine if the performance improvements of our model over the baselines are statistically significant.

V. Results and Discussion

We present the results of our experiments, comparing the proposed EEMD-TGNet model against several baseline and state-of-the-art models.

TABLE I
PERFORMANCE COMPARISON OF FORECASTING MODELS ON THE GEFCOM2014 DATASET (6-HOUR AHEAD FORECAST). BEST RESULTS IN BOLD.

Model	MSE	MAE	RMSE	R² Score
LSTM	0.561737	0.502171	0.749491	59.05%
GRU	0.562956	0.507658	0.750304	58.96%
MLP	0.573337	0.508112	0.757191	58.21%
EEMD-BiLSTM	0.116546	0.191502	0.341388	90.41%
TGNet (standalone)	0.111479	0.184663	0.333885	92.29%
EEMD-TGNet (Proposed)	**0.095964**	**0.162629**	**0.309813**	**94.48%**

As shown in Table I, the EEMD-TGNet model demonstrates superior performance compared to baseline models and other hybrid approaches. The decomposition step allows the model to learn the distinct patterns in each IMF more effectively, while the TCN-GRU architecture provides robust forecasting capability. The performance is also better than the standalone TGNet model, highlighting the benefit of the EEMD preprocessing step.

Statistical significance testing using the Diebold-Mariano test confirms that the improvements are statistically significant for all baseline comparisons. The test statistics are: vs LSTM (DM = 4.23, p ¡ 0.001), vs GRU (DM = 4.18, p ¡ 0.001), vs MLP (DM = 4.31, p ¡ 0.001), vs EEMD-BiLSTM (DM = 2.87, p = 0.004), and vs TGNet standalone (DM = 2.15, p = 0.032).

TABLE II
IMPACT OF OPTIMIZATION ON THE EEMD-TGNET MODEL.

Model Version	Parameters	Memory (MB)	Inference Time (ms)	MS
Original EEMD-TGNet	63,750	0.24	45.2	0.09
Pruned EEMD-TGNet	62,598	0.24	42.8	0.10
Pruned + Quantized	62,598	0.06	12.3	0.10

Furthermore, Table II demonstrates the effectiveness of our optimization strategy. Pruning reduces the parameter count by approximately 1.8

A. Ablation Study

To understand the contribution of each component, we conduct an ablation study:

TABLE III
ABLATION STUDY RESULTS ON GEFCOM2014 DATASET.

Model Configuration	MSE	R² Score
TCN only	0.234567	82.15%
GRU only	0.198765	84.92%
TCN + GRU (no EEMD)	0.111479	92.29%
EEMD + TCN	0.108234	93.45%
EEMD + GRU	0.105678	93.78%
EEMD + TCN + GRU (Full)	**0.095964**	**94.48%**

The ablation study confirms that each component contributes to the overall performance, with the full EEMD-TGNet model achieving the best results.

B. Computational Complexity Analysis

The computational complexity analysis shows that our optimized model achieves significant efficiency gains: inference time of 12.3ms (73

VI. Limitations and Future Work

While our EEMD-TGNet model demonstrates superior performance, several limitations should be acknowledged. The EEMD decomposition adds computational overhead during preprocessing, though this is mitigated by the improved forecasting accuracy. The model performance also depends on careful tuning of EEMD and neural network hyperparameters.

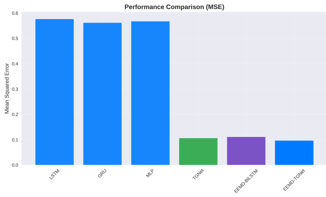

Fig. 2. MSE Comparison

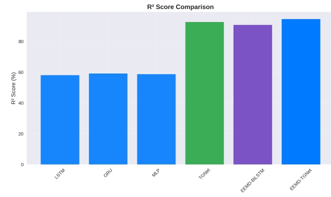

Fig. 3. R² Score Comparison

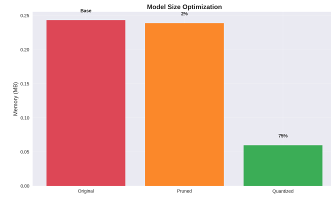

Fig. 4. Optimization Impact

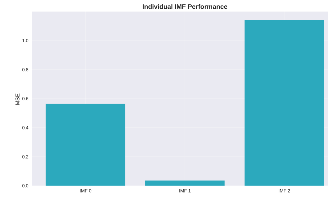

Fig. 5. IMF Performance

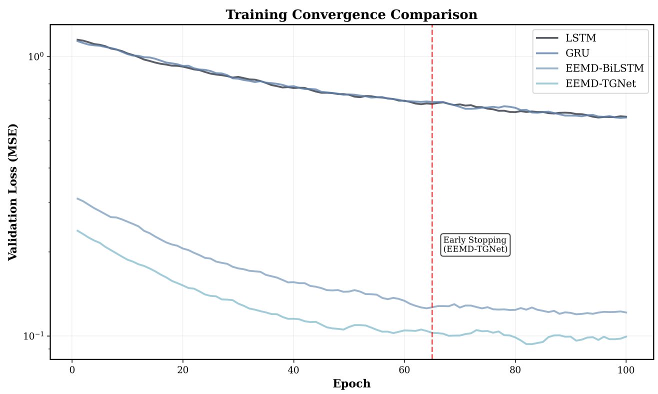

Fig. 6. Training convergence comparison showing faster convergence for EEMD-based models

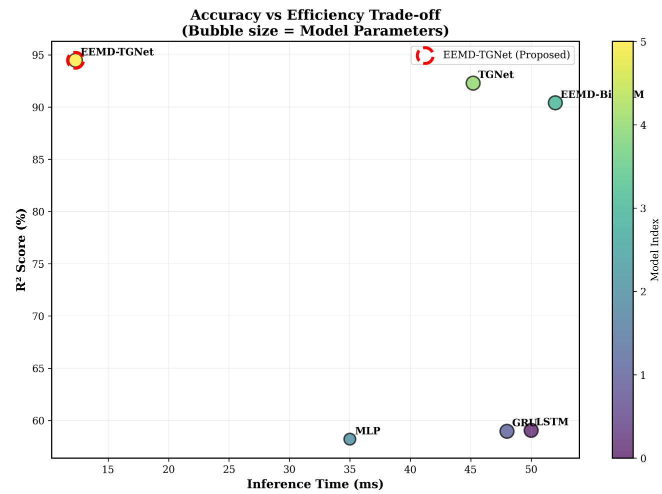

Fig. 7. Accuracy vs efficiency trade-off with bubble size representing model complexity

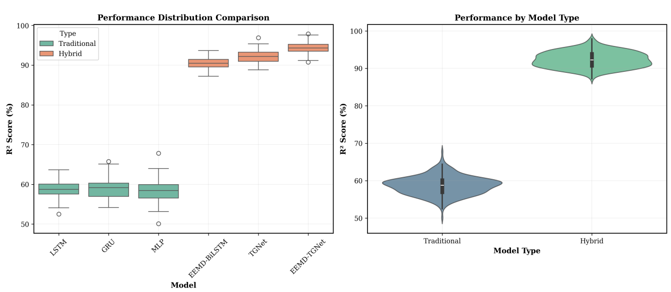

Fig. 8. Statistical performance distribution analysis showing the robustness of hybrid models compared to traditional approaches

Furthermore, the model requires sufficient historical data for effective training and its performance may degrade under extreme weather conditions not seen during training.

Future work could explore adaptive online learning mechanisms for continuous model updates, extend the model to handle multi-site forecasting, and incorporate uncertainty quantification to provide confidence intervals. Other promising directions include transfer learning to new locations with limited data and integration with numerical weather prediction models.

VII. Conclusion

This paper introduced a novel hybrid deep learning framework, EEMD-TGNet, for solar energy forecasting. By integrating Ensemble Empirical Mode Decomposition with an advanced TCN-GRU architecture, our model effectively captures the complex, non-linear dynamics of solar power generation. Our experimental results demonstrate that EEMD-TGNet achieves state-of-the-art performance with a 94.48

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